

AI STUDYBUDDY API

Code-Layout, Readability and Reusability

Observed source layout and code review

TEAM ID	SWTID-2026-2776
TEAM SIZE	5
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Layout

index.js; src/controllers/{authController,materialController,adminController}.js; src/middleware/{auth,upload}.js; src/models/{User,Material}.js; src/routes/{auth,materials,admin}.js; src/utls/{db,gemini,tokens}.js.

Separation

Routes map endpoint paths; controllers coordinate persistence and AI; models shape data; auth middleware verifies tokens and role; upload middleware configures Multer; utls isolate database/token/AI helpers.

Maintainability risks

AI endpoint ownership checks are inconsistent; upload parsing treats every accepted file as UTF-8 text; flashcard/quiz JSON parsing assumes valid model output; admin role can be requested during registration.

Recommended improvements

Use PDF text extraction, restrict role assignment, check resource ownership on every material action, validate IDs/body limits, handle temporary-file cleanup on errors, and add automated tests.